

Quantifying the Operational Benefits of Deep Learning based Dynamic Traffic Prediction using Real-World Dataset

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Abstract: A convolutional neural network is trained using real-world data, for dynamic prediction of the required transceivers supporting 6G X-haul, leading to 20% and 16% lower average transceiver utilization over static and semi-static cases, respectively. © 2024 The Author(s)

1. Introduction

Dynamic traffic prediction in X-haul networks offers significant operational advantages, including improved resource allocation efficiency and lower power consumption compared to static resource allocation. By utilizing Deep Learning (DL), the number of required transceivers is estimated on a per-hour basis more accurately than simply averaging over past data, when benchmarked against actual needs. At the same time, the risk of overutilization is reduced, even during fluctuating traffic conditions. In prior art, Machine Learning (ML) methods have been exploited for traffic prediction [1]-[3]. The most accurate ones were the Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and a mix of the two. In this work, we exploit a custom deep CNN, due to its abilities to capture the spatial dependencies in traffic patterns and to automatically extract relevant features. Our deep CNN is trained on a real-world dataset collected between 2022 and 2023 from a national carrier. To the best of our knowledge, this is the most recent dataset that contains 4/5G data, allowing us to study optimization means of the X-haul, using the dimensions of a real network and extracting generalized conclusions about the benefits of dynamic transceiver allocation over static and semi-static cases, while ensuring that minimal overutilization occurs. This is also the first work that estimates the possible energy savings in the X-haul, based on real data, as the selected transceivers utilize switch-off / standby mode capabilities. More specifically, the X-haul includes multiple XR400 transceivers [4], with subcarrier multiplexing capabilities, each employing 16 subcarriers in conjunction with our novel “Spatially-Diverse Point-to-Multi-Point (SDPtMP) node” element which can be shared across multiple BSs. This low-loss element provides an increased power budget compared with a Passive Optical Network (PON), due to the reduction of splitters, further contributing in power savings coming from the lower optical output power requirements from the XR400 transceivers.

2. X-haul traffic and key assumptions

2.1 The midhaul network

The X-haul network under study, depicted in Fig.1, consists of three segments: the fronthaul, the midhaul, and the backhaul. The midhaul network architecture in Telefónica’s infrastructure typically comprises hierarchical nodes in horseshoe ring structures that aggregate traffic from the different locations of the metro and access domains. Each node typically serves 50 to 100 Base Stations (BSs) and the midhaul ring comprises between 1 and 9 nodes. We place our novel SDPtMP element [5] on each node, which inserts four parallel feeds to the fronthaul segment, allowing a single transceiver to serve up to four BSs. The SDPtMP node allocates three subcarriers to each BS through subcarrier interlacing, using passive filtering techniques to distribute the parallel feeds across fronthaul fiber feeds. As shown in inset (ii) of Fig.1, the SDPtMP node is the gateway to the fronthaul segment, with four parallel fibers (Space Division Multiplexing - SDM ingress/egress), each capable of carrying WDM traffic in subcarrier groups. The interlacing process is implemented by low-loss sharp optical interleaves arranged in two stages, interlacing I/O ports at the sub-channel level. Using a 64 GHz periodicity, the first stage produces complementary 30 GHz wide channels. The second stage, using the same periodicity but shifted by $\frac{1}{4}$ cycle, results in four distinct subcarrier groups (corresponding to one out of four inputs), each 14 GHz wide, sufficient to support three subcarriers of 4 GHz bandwidth each with small margin. We should mention that the same wavelength can appear on multiple inputs, making the SDPtMP node wavelength contentionless. It is recommended to first use the same wavelength across all spatial inputs before introducing additional wavelength channels. This allows the receiver to detect all four incoming spatial channels via fixed subcarrier associations, effectively quadrupling the incoming capacity compared to traditional PON architectures. Each node is equipped with XR400 prototype transceivers [4], capable of generating up to 16 independent subcarriers, each operating at approximately 4 GBaud and utilizing dual-polarization 16-QAM modulation. This configuration delivers a net data rate of 25 Gb/s per subcarrier and 75 Gb/s

per BS. Using the above considerations, the total traffic of a midhaul ring of average size (5 nodes) that serves on average 75 BSs, each with a rate of 75 Gb/s, leads to 28.125 Tb/s. This traffic can be supported by 95 transceivers.

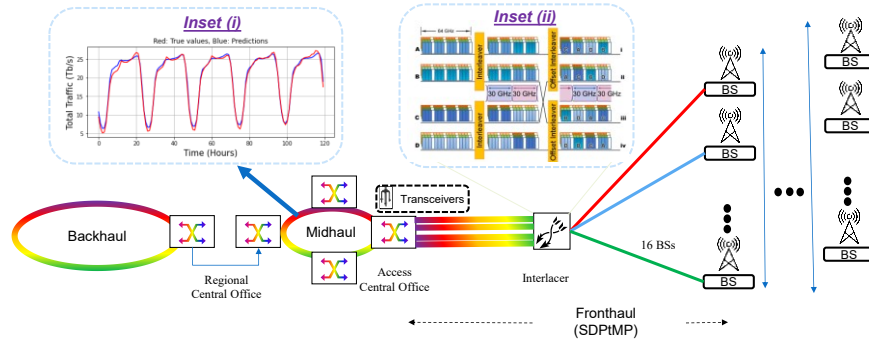


Fig. 1. The network under study

2.2 Details of the dataset collected from a real network

The dataset under study contains real data collected during 2022 and 2023 from Telefónica's Network in a city in South America, representing the total traffic of 600k subscribers at four network gateways. The data spans over 358 days, and each day contains 24 measurements (one per hour), leading to a total of 8592 values. Maximum traffic is observed between 19:00 and 22:00, and the lowest traffic between 1:00 and 4:00. An example of the traffic pattern over a five-day period is illustrated in inset (i) of Fig.1 (red curve). The dataset contains mainly 4G (and a small fraction of 5G) traffic. To scale the traffic to the midhaul capabilities, we make the following assumptions. First, our target is a city of more significant scale, e.g. 3M subscribers, leading to a 5-times multiplication factor. Next, by scaling the 4G peak capacity of 1 Gb/s [6] to the peak capacity of 6G, which reaches 400 Gb/s for the long-term evolution scenario [7], there is also another 400 times multiplication factor. These two factors increase the maximum traffic by 2,000 times, scaling the traffic during the peak hours from about 14.05 Gb/s to 28.1 Tb/s, which is the estimated traffic for the average midhaul scenario presented in the previous section.

2.3 Deep Learning for traffic estimation

To predict the traffic by recognizing patterns and extracting features automatically from the data, we employed a custom lightweight deep CNN (of 33,569 parameters) with five layers. The first layer included 16 convolutional filters with a size of 96, followed by a ReLU activation function, a 4-strided max-pooling operation, and a dropout probability equal to 0.25. The second layer incorporated 32 convolutional filters with a size of 24, followed by a ReLU, a 4-strided max-pooling operation, and a dropout probability equal to 0.25. The third layer incorporated 48 convolutional filters with a size of 6, followed by a ReLU, a 3-strided max-pooling operation, and a dropout probability equal to 0.25. The fourth layer incorporated 64 convolutional filters with a size of 2, followed by a ReLU, a 2-strided max-pooling operation, and a dropout probability equal to 0.25. The final layer contained 64 hidden units, followed by a linear activation function. For the training, we divided the dataset into train, validation, and test subsets. The data from the first 212 days formed the test set, the next 68 days the validation set and the final days the test set. Each input sample has 96 values (data of 4 consecutive days), and the CNN's output was the traffic for the next hour. An overlap of 95/96 was considered as an attempt to estimate the traffic of all possible hours in the test set as well as to create more training samples. Before being injected into the CNN, the data were normalized using Z-score normalization. Finally, Adam was the optimizer, the maximum number of epochs was set to 500 and the training stopped if the loss in the validation set did not improve after 50 epochs. The predicted traffic pattern (blue curve) over a five-day period in comparison with the actual traffic (red curve) is illustrated in inset (i) of Fig.1.

3. Results

In this section, we estimate the number of transceivers required on an hourly basis in the test set for a) a static case, considering that all 95 transceivers are exploited during all hours and days, b) a semi-static case, considering that the number of transceivers is adjusted every 8 hours, according to the average value of these 8 hours and scaled up to ensure zero under-allocation in the validation set, c) a dynamic case, calculating the number of transceivers on an hourly basis using the CNN, and slightly scaling it up to about 15% to avoid resource under-allocation in the validation set. Figure 2 shows the calculated number of transceivers on an hourly basis for the three cases. The dynamic case illustrates the number of transceivers averaged over all days in the test set for the different hours. The first observation is that the dynamic case leads to a reduced number of transceivers, especially for the hours between 23:00 and 6:00, where the requirements are minimized. However, even in the hours 6:00 to 18:00, the number of

transceivers is between 2.5% and 22% lower than the peak hour. The semi-static case provides 10% lower transceivers during off-peak hours than the static case, leading to a 5% lower average number of transceivers over the entire day. Overall, the dynamic case leads to an average of about 20% lower number of transceivers than the static case and 16% lower number of transceivers than the semi-static case. These gains can be directly translated to up to a total of 20% lower power consumption in case of transceiver switch off. However, in low power and standby modes, the power consumption gains are expected to be smaller. The work is ongoing to quantify these gains.

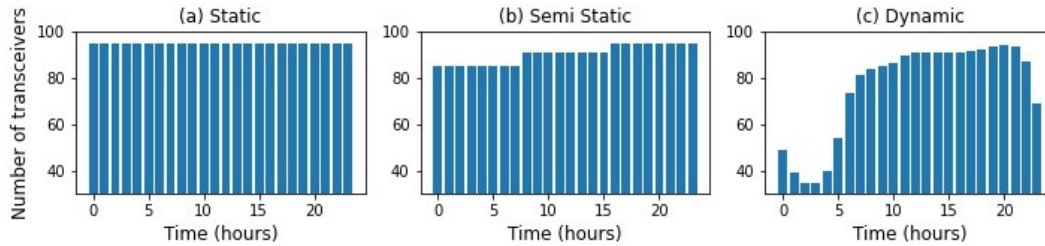


Fig. 2. Number of operational transceivers estimated per hour for the three resource allocation cases.

The second set of results illustrated in Fig.3 shows the percentage of overallocated transceivers. This compares the number of transceivers for the three cases against the actual needs on the test set, when estimated for all 24 hours. As expected, there is a high overallocation in all three cases, especially in off-peak hours (between 01:00 and 06:00). This happens because, during these hours, the traffic variance is the highest, making the predictions harder than the peak hours. This, in turn, increases the number of assigned transceivers to avoid any potential under-allocation which can negatively impact the Quality of Service (QoS). Overall, the average over-allocation over all days of the test set for the static, semi-static and dynamic cases is 92.5%, 86.7% and 28.8%, respectively, demonstrating the superiority of the dynamic case, especially during off-peak hours.

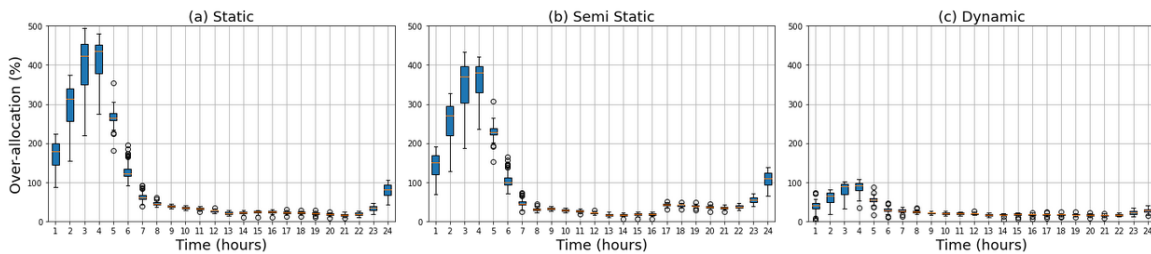


Fig. 3. Over-allocated transceivers [%] compared with the true needs for the three operational cases.

4. Conclusions

This work demonstrated that exploiting DL on real-world datasets allows the dynamic approach to reduce transceiver requirements by 20% compared to static cases and 16% compared to semi-static cases. These reductions directly translate into reduced power consumption of up to 20%, when transceiver switch off is considered. Additionally, this approach effectively addresses the challenges of resource overallocation during off-peak hours, ensuring that service quality and network performance are maintained while preventing potential resource over-utilization. A future study will examine the impact of transceivers with the capability to flexibly allocate the number of subcarriers to BSs according to their demands. Finally, we plan to perform traffic prediction studies for various system parameters/details to feed digital twins for the optimal functional split selection to match the users demands.

Acknowledgement

This work was supported by the EU's Horizon Europe program under the PROTEUS-6G project under Grant Agreement 101139134.

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